

Divergence Identities for Probabilistic Goal Recognition

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Abstract

Goal recognition aims to infer an agent’s goal from partial observations of its behaviour. Existing approaches derive likelihoods from the costs of plans consistent with observations, often relying on trajectory-level reasoning and Boltzmann-rational models. In this work, we adopt a distributional perspective in which both goals and observations are represented as distributions over an outcome domain. We show that likelihoods and posterior probabilities can be characterised through identities that relate GR to divergences between observation-induced and goal-intended distributions. These identities expose a common structure across inference regimes, unifying probabilistic (sum-product) and optimisation-based (max-product) formulations. Our formulation captures cost-based goal recognition as a special case, and provides a common perspective alongside related approaches such as inverse reinforcement learning and Boltzmann-rational models of behaviour. By deriving existing models in a more general probabilistic setting, the framework offers an abstract account of goal recognition and formally situates known results within a broader class of inference problems over structured distributions.

Introduction

Goal recognition aims to infer an agent’s goal from partial observations of its behaviour. This problem is fundamental in a wide range of settings, including human-aware planning, assistive systems, and monitoring of autonomous agents. Early approaches to goal and plan recognition rely on matching observations against predefined plan or policy libraries, where recognition is performed by identifying plans that are consistent with the observed behaviour (Kautz and Allen 1986; Charniak and Goldman 1993). While effective in restricted settings, such approaches are inherently limited by the coverage of the library and lack a principled generative interpretation of observations.

More recent work addresses these limitations by formulating goal recognition as a planning problem, deriving likelihoods from the relative costs of plans that achieve candidate goals and are consistent with the observations (Ramirez and Geffner 2009, 2010). In this framework, observations are treated as constraints over trajectories, and the likelihood of a goal is estimated from the cost of optimal or near-optimal plans that satisfy these constraints. A complementary line of work retains the cost-difference intuition, but derives the

relevant costs from learned behavioural models rather than from a planning domain. For instance, Polyvyanyy et al. (2020) use process mining to discover goal-specific process models from traces, and infer goals by measuring deviation costs between a new observation and each discovered model.

Parallel results can be found in Inverse Reinforcement Learning (IRL) (Ng, Russell et al. 2000), which provides a different but related perspective to solve a similar problem. It addresses the problem of explaining observations through latent objective functions, inferring underlying cost or reward structures from histories of an acting agent’s behaviour. In its maximum entropy formulation (Ziebart et al. 2008), IRL models behaviour as a Boltzmann distribution over trajectories, $P(\tau \mid \theta) \propto e^{-\beta c_\theta(\tau)}$, where β controls the degree of rationality, trading off optimality and stochasticity. This formulation captures the intuition that lower-cost trajectories are more likely, and provides a principled probabilistic framework for inferring objective functions that explain observed behaviour. Similarly, related work in robotics draws on maximum-entropy IRL to define a cost-based distribution over trajectories, and performs goal inference from partial trajectories by marginalising over trajectory completions (Dragan, Lee, and Srinivasa 2013). In these approaches, observations are initial trajectories, and inference sums over completions consistent with them.

We adopt a distributional perspective to GR in which candidate goals and observations are associated with distributions over an outcome domain (e.g. plans, trajectories). In this setting, inference is characterised through identities relating likelihoods and posterior probabilities to divergences between observation-induced and goal-intended distributions. These identities expose common structure across inference regimes: posterior inference decomposes into divergence-based compatibility and residual terms, with analogous probabilistic- (sum-product) and optimisation-based (max-product) forms. In line with established intuition, we show that our sum-product and max-product formulas align under increasingly rational regimes, and derive key properties of existing cost-based models when instantiated with cost scores.

Our formulation is not a new isolated model, but a unifying framework that subsumes existing approaches such as cost-based goal recognition, and offers a common perspective that aims to facilitate connections to models such

as inverse reinforcement learning. By explicitly introducing an outcome mapping and deriving both sum-product and max-product inference from a shared divergence-based viewpoint, the framework explores how prior cost-based goal-recognition models arise from a broader probabilistic setting. Expressing likelihoods and posteriors through explicit measures of compatibility between distributions over observations and candidate goals provides an interpretable account of goal recognition, and situates goal recognition more precisely as an instance of inference over structured distributions.

Divergence Identities

Probabilistic setting. Goal recognition is the problem of inferring an agent’s intended goal from partial observations of its behaviour. We consider a setting with an underlying joint distribution over outcomes and the candidate goals associated with them, while observations reveal only partial information about these outcomes. Let $\mathcal{Z}$ denote a discrete mutually exclusive outcome space, let $G \in \mathcal{G}$ be a candidate goal. The space $\mathcal{Z}$ may itself represent an abstraction of a more detailed latent process. Let $y^* \in \mathcal{Y}$ denote an observation. We assume a deterministic compatibility relation

$$\sim \subseteq \mathcal{Z} \times \mathcal{Y}$$

where $z \sim y^*$ means that outcome z is compatible with the observation y^* ; in other words, a key assumption is that

$$\rho(y^* | z) = \mathbf{1}[z \sim y^*].$$

This occurs because we assume a setting where observations are deterministic given the outcome: once z is fixed, y^* is either compatible with it or not. We write $\rho(y^* | G)$ for the probability mass, under G , of outcomes compatible with y^* . Thus, an observation may describe only part of the underlying outcome, such as selected features, a partial trajectory, a set of facts achieved along a plan.

Such probabilistic setting, and the scope of our work, thus aligns with the broad set of deterministic partial-observation goal recognition problems with the key assumption of *deterministic compatibility*: a complete outcome either is or is not compatible with the observed behaviour.

It follows that this deterministic observation setting entails another key simplification for our results,

$$\rho(y^* | z, G)\rho(z | G) = \mathbf{1}[z \sim y^*]\rho(z | G), \quad (1)$$

since when z is fixed, whether y^* is observed is determined solely by the compatibility relation $z \sim y^*$. This is because z represents the underlying outcome: the observation model only checks whether this outcome entails the observed description. The goal G may affect which outcomes are likely, through $\rho(z | G)$, but it does not change what observation a fixed outcome supports. Thus the observation is conditionally independent of the goal given the outcome. If z is impossible under G , then this is handled by $\rho(z | G) = 0$, so the product $\rho(y^* | z, G)\rho(z | G)$ is zero regardless.

Example 1. Let outcomes $z \in \mathcal{Z}$ be complete action sequences $z = \langle a_1, \dots, a_n \rangle$, and let the observation be a partial action sequence $y^* = \langle o_1, \dots, o_m \rangle$. We say that

z is compatible with y^* , written $z \sim y^*$, iff z embeds the observation: there exists a strictly increasing map $f : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ such that $a_{f(j)} = o_j$ for every j . Thus y^* need not be a contiguous subsequence of z , but its actions must occur in the same order.

Under deterministic compatibility, once the complete action sequence z is fixed, there is no remaining uncertainty about whether the partial observation y^* is present in it. The observation is obtained simply by checking whether y^* is embedded in z . Therefore, $\rho(y^* | z) = \mathbf{1}[z \sim y^*]$.

If $\rho(z | G) > 0$, then z is a possible outcome under candidate goal G . Conditioning on G may change the probability assigned to z , but once this outcome is fixed, whether it embeds y^* is already determined by the sequence itself. If $\rho(z | G) = 0$, then z contributes no mass under G , so both sides of the product are zero. Hence $\rho(y^* | z, G)\rho(z | G) = \mathbf{1}[z \sim y^*]\rho(z | G)$.

Discrete Sum–Product Identities

Given $\mathcal{Z}, G \in \mathcal{G}$ and y^* , let $\rho(z | G)$ be the goal-conditioned distribution over outcomes, and let $\rho(z | y^*, G)$ be the corresponding distribution updated by the observation y^* .

Theorem 1 (Likelihood identity).

$$D_{\text{KL}}(\rho(z | y^*, G) \| \rho(z | G)) = -\log \rho(y^* | G).$$

Proofs are included at the end of the paper. This identity shows that the likelihood of an observation under a goal can be interpreted as the compatibility between the goal-conditioned outcome distribution $\rho(z | G)$ and the distribution induced after conditioning on the observation, $\rho(z | y^*, G)$. Equivalently, $\rho(y^* | G)$ is the exponential of the negative KL divergence between these distributions. Thus, observations that induce only a small change to the goal-conditioned distribution are more likely under that goal. This parallels cost-Boltzmann goal recognition, where likelihoods are exponential in negative costs, but replaces the cost term with a divergence between outcome distributions.

Theorem 2 (Posterior identity).

$$-\log \rho(G | y^*) = D_{\text{KL}}(\rho(z | y^*, G) \| \rho(z | y^*)) + \mathbb{E}_{\rho(z | y^*, G)} [-\log \rho(G | z, y^*)].$$

The divergence term compares the outcome distribution induced by conditioning on both the observation and the candidate goal with the outcome distribution induced by the observation alone. It therefore measures how much the goal changes the distribution over latent outcomes after y^* has already been observed. A small divergence indicates that the candidate goal is compatible with the observation-induced outcome distribution, while a larger divergence indicates that explaining the observation under that goal requires a more substantial shift in the posterior over outcomes.

The expectation term in the posterior identity accounts for residual goal ambiguity. When outcomes only achieve non-overlapping goals, observing the latent outcome z determines the goal (i.e., $\rho(G | z, y^*) = \mathbf{1}[z \models G]$), so on the support of $\rho(z | y^*, G)$, this term vanishes and the posterior is given exactly by the divergence term. When goals

overlap, however, the same outcome may remain compatible with multiple goals. In this case, the expectation over outcomes consistent with y^* captures the remaining uncertainty about which goal is intended, correcting the posterior beyond what the divergence term alone can represent.

Example 2. Let z be an abstract outcome represented only by the set of facts achieved by some underlying plan. For instance, suppose

$$z = \{p, q\}$$

means that the unknown underlying plan achieves facts p and q , but does not identify which concrete plan achieved them. Let $y^* = \{p\}$ be an observed fact set, so that $z \sim y^*$ whenever $y^* \subseteq z$.

Because $\mathcal{Z}$ abstracts away from the underlying set of concrete plans, the same fact set may be compatible with multiple goals. For example, plans achieving $\{p, q\}$ may achieve goal G_1 with probability 0.7 and goal G_2 with probability 0.3. Then observing the abstract outcome z does not determine the intended goal, so

$$0 < \rho(G_1 | z) < 1.$$

The expectation term in Theorem 2 captures exactly this residual goal ambiguity: even after conditioning on the outcome abstraction z , some uncertainty about the intended goal remains.

Example 3. If instead z denotes a concrete underlying plan rather than its achieved fact abstraction, goal achievement may be deterministic. Suppose every concrete plan achieves exactly one candidate goal. Then, for each z ,

$$\rho(G | z) = \mathbf{1}[z \models G].$$

On the support of $\rho(z | y^*, G)$, we have $z \models G$, and therefore $-\log \rho(G | z, y^*) = 0$. Hence the expectation term in Theorem 2 vanishes, leaving the posterior determined solely by the divergence term.

Relation to planning-as-goal-recognition. A practical application of this model is provided by *Planning as Goal Recognition* (Rosa et al. 2026), where goal-recognition models are used to derive heuristic guidance for classical planning search. The key connection is that, rather than deriving likelihoods from plan-cost differences, this approach directly estimates goal-conditioned likelihood estimates for observing facts, similar to Example 2, providing a practical application of Theorem 1. Through the divergence identities, these scores can be interpreted as compatibility measures between an estimated goal-conditioned distribution over facts and its observation-conditioned counterpart. This gives a distributional explanation for why such scores can guide search toward states that appear intended towards the planning task’s goal.

Discrete Max-Product Identities

Max-product factorisation. We adopt a max-product formulation, where summation is replaced by maximisation: the sum-product model uses the semiring $(\mathbb{R}_{\geq 0}, +, \cdot)$; the max-product analogue replaces summation by maximisation: $(\mathbb{R}_{\geq 0}, \max, \cdot)$, or, in log-cost space, $(\mathbb{R} \cup$

$\{\infty\}, \min, +)$. In this setting, this means that joint probabilities factorise multiplicatively, but marginalisation is defined by maximisation. Technically, we are not talking about probabilities anymore, but rather optimisation-based scores, and probability distributions ρ become max-product score distributions $\rho_{\max}$, where the normalising factor is obtained through a max-marginal as opposed to a sum-marginal.

This max-product approach is closely related to the qualitative probability perspective of Goldszmidt and Pearl (1996) which informs the order-of-magnitude interpretation used by Ramirez and Geffner (2010),

$$P(y^* | G) = \sum_{\pi: \pi \sim y^*} P(\pi | G) \approx \max_{\pi: \pi \sim y^*} P(\pi | G),$$

to justify their cost-based likelihood by approximating a sum over compatible plans with its dominant term.

In contrast to treating this dominance assumption only as an approximation to a sum-product likelihood, we use it as the underlying Bayesian inference regime: marginalisation, conditioning, likelihoods, and posteriors are all defined consistently in max-product space. Thus, for two variables x and y , marginals are defined as

$$\rho_{\max}(y) := \max_x \rho_{\max}(x, y)$$

and conditionals are defined

$$\rho_{\max}(x | y) := \frac{\rho_{\max}(x, y)}{\rho_{\max}(y)}$$

We can obtain a max-product version of Bayes’ rule:

$$\rho_{\max}(x, y) = \rho_{\max}(x | y) \rho_{\max}(y) = \rho_{\max}(y | x) \rho_{\max}(x),$$

$$\rho_{\max}(x | y) = \frac{\rho_{\max}(y | x) \rho_{\max}(x)}{\max_x \rho_{\max}(y | x) \rho_{\max}(x)}.$$

Max-product likelihood. As previously, $\rho_{\max}(z, y^*) = \rho_{\max}(z)$ if z is compatible with y^* , and 0 otherwise; thus, $\rho_{\max}(y^* | z) = \mathbf{1}[z \sim y^*]$. Using max-product distribution $\rho_{\max}$ with max-marginalisation, the max-marginals for the goal are

$$\rho_{\max}(G, y^*) = \max_z \rho_{\max}(G, z, y^*),$$

$$\rho_{\max}(y^*) = \max_{G, z} \rho_{\max}(G, z, y^*).$$

and we have that

$$\rho_{\max}(z | G) = \frac{\rho_{\max}(z, G)}{\rho_{\max}(G)}.$$

The max-normalised conditional score becomes

$$\rho_{\max}(z | y^*, G) = \frac{\rho_{\max}(y^* | z, G) \rho_{\max}(z | G)}{\rho_{\max}(y^* | G)}$$

Following the same reasoning as Equation 1, we still have that whenever $\rho_{\max}(z | G) > 0$, then $\rho_{\max}(y^* | z, G)$ is determined by whether $z \sim y^*$, thus obtaining

$$\rho_{\max}(z | y^*, G) = \frac{\mathbf{1}[z \sim y^*] \rho_{\max}(z | G)}{\rho_{\max}(y^* | G)}.$$

We define the max-divergence

$$D_{\max}(q \| p) := \max_{z: q(z) > 0} \log \frac{q(z)}{p(z)}.$$

Theorem 3 (Max-product likelihood identity).

$$D_{\max}(\rho_{\max}(z | y^*, G) \| \rho_{\max}(z | G)) = -\log \rho_{\max}(y^* | G).$$

Max-product posterior. From the max-product Bayes relation we can also get

$$\rho_{\max}(z \mid y^*, G) = \frac{\rho_{\max}(G \mid z, y^*) \rho_{\max}(z \mid y^*)}{\rho_{\max}(G \mid y^*)}.$$

Theorem 4 (Max-product posterior identity).

$$-\log \rho_{\max}(G \mid y^*) = D_{\max}(\rho_{\max}(z \mid y^*, G) \parallel \rho_{\max}(z \mid y^*)) + \min_z [-\log \rho_{\max}(G \mid z, y^*)].$$

These identities are the max-product analogues of the preceding KL decompositions, with likelihoods and posteriors interpreted as max-normalised scores rather than ordinary probabilities. The likelihood identity again interprets the observation score under a goal as compatibility between the goal-conditioned outcome score and the score obtained after conditioning on the observation, but with KL divergence replaced by the max-divergence induced by optimisation over highest-scoring explanations. Similarly, the posterior identity preserves the same structure as the sum-product case: the max-posterior score decomposes into a divergence-based compatibility term and a residual term. The difference is that expectations are replaced by minima, reflecting the shift from marginalising over possible outcomes to selecting the best explanation.

Deriving Boltzmann cost-based models. As an instantiation of the max-product model, and following the cost-based view of Ramirez and Geffner (2010), we associate more optimal outcomes with higher scores. Under the max-product formulation above, costs induce optimisation-based scores over latent outcomes, while observations and goals act as constraints over admissible explanations. For this reason, we interpret the resulting likelihoods and posteriors as max-product analogues of probabilistic quantities rather than ordinary sum-normalised probabilities, naturally yielding the cost-gap structure central to cost-based goal recognition.

Let outcomes carry intrinsic costs $c : \mathcal{Z} \rightarrow \mathbb{R}$, while goals and observations act as deterministic constraints over outcomes. We start from the max-product joint score

$$\rho_{\max}(z, y^*, G) := \mathbf{1}[z \sim y^*] \mathbf{1}[z \models G] e^{-\beta c(z)}.$$

Here, $c(z)$ is the intrinsic cost of outcome z , while $z \sim y^*$ and $z \models G$ determine whether the outcome is compatible with the observation and achieves the candidate goal. Thus, the goal does not change the cost of an outcome; it only restricts which outcomes are admissible. Conditioning on a candidate goal G gives

$$\begin{aligned} \rho_{\max}(z \mid G) &= \frac{\mathbf{1}[z \models G] e^{-\beta c(z)}}{\max_{\bar{z}: \bar{z} \models G} e^{-\beta c(\bar{z})}} \\ &= \mathbf{1}[z \models G] e^{-\beta(c(z) - c^*(G))}, \end{aligned}$$

where

$$c^*(G) := \min_{\bar{z}: \bar{z} \models G} c(\bar{z}).$$

Thus, once G is assumed, each outcome is evaluated by its excess cost relative to the optimal outcome for the same

goal. This matches the goal recognition view that observations are informative when they are close to optimal under a candidate goal, rather than when the candidate goal itself is cheap in absolute terms.

Here, $\mathbf{1}[z \models G]$ reflects the plan-cost abstraction, corresponding to the simplifying assumption $\rho_{\max}(G \mid z) = \mathbf{1}[z \models G]$. Since this scalar-cost abstraction represents explanations only by their cost once goal achievement is fixed, it loses information about whether the same underlying plan also supports other candidate goals. Under this abstraction, we obtain the following cost-based max-product forms.

Corollary 1 (Cost-Boltzmann Likelihood).

$$\rho_{\max}(y^* \mid G) = e^{-\beta(c^*(y^*, G) - c^*(G))},$$

where

$$c^*(y^*, G) := \min_{z: z \sim y^*, z \models G} c(z), \quad c^*(G) := \min_{z: z \models G} c(z).$$

The likelihood compares the cheapest explanation compatible with both the observation and goal against the cheapest explanation achieving the goal at all. Hence the relevant quantity is the excess cost

$$\Delta(G, y^*) := c^*(y^*, G) - c^*(G).$$

Corollary 2 (Cost-Difference Posterior with External Prior). *For an externally specified max-product prior $\pi_{\max}(G)$ over candidate goals,*

$$\rho_{\max}(G \mid y^*) = \frac{e^{-\beta \Delta(G, y^*)} \pi_{\max}(G)}{\max_{G'} e^{-\beta \Delta(G', y^*)} \pi_{\max}(G')}.$$

In particular, under a uniform max-product prior over candidate goals, the prior terms cancel and the posterior reduces to

$$\rho_{\max}(G \mid y^*) = e^{-\beta(\Delta(G, y^*) - \min_{G'} \Delta(G', y^*))}.$$

This is the usual cost-difference posterior: goals are compared by how much additional cost is required to explain the observation once each goal is assumed. It is appropriate when no candidate goal is preferred a priori in the problem's initial state, or when prior goal preferences are supplied independently of the costs of achieving the goals.

Corollary 3 (Posterior Induced by the Same Cost Model). *If no external prior is introduced, the intrinsic joint score induces a max-product prior over goals by max-marginalising over outcomes: $\rho_{\max}(G) = \max_{\bar{z}: \bar{z} \models G} e^{-\beta c(\bar{z})} = e^{-\beta c^*(G)}$. Combining this induced prior with the cost-difference likelihood gives*

$$\rho_{\max}(G \mid y^*) = \frac{e^{-\beta c^*(y^*, G)}}{\max_{G'} e^{-\beta c^*(y^*, G')}}.$$

This second posterior is the absolute-cost posterior. It arises because, under the intrinsic joint score, goals are treated as properties of outcomes: if a goal can be achieved by a shorter outcome, then it receives a larger prior score before the observation is considered. Consequently,

$$e^{-\beta(c^*(y^*, G) - c^*(G))} e^{-\beta c^*(G)} = e^{-\beta c^*(y^*, G)},$$

so the goal baseline $c^*(G)$ cancels. The posterior then ranks goals by the absolute cost of the cheapest outcome that both explains the observation and achieves the goal.

The distinction between Corollary 2 and Corollary 3 is therefore not a contradiction, but a modelling choice. A cost-difference posterior is obtained when the likelihood is combined with a uniform or externally specified goal prior, as is typically the case in GR. If instead the prior is induced from the same outcome-cost model, then cheaper goals are already more likely a priori, yielding the absolute-cost posterior.

The external-prior choice can alternatively be encoded directly in a modified joint-style score which more accurately reflects common GR modelling choices,

$$\tilde{\rho}_{\max}(z, y^*, G) := \mathbf{1}[z \sim y^*] \mathbf{1}[z \models G] e^{-\beta(c(z) - c^*(G))} \pi_{\max}(G),$$

whose max-marginal over z gives the cost-difference posterior numerator directly. The modified score should be understood as a way of encoding the external-prior posterior directly at the outcome level. It does not make the cost of an outcome goal-dependent: $c(z)$ remains the intrinsic outcome cost. Instead, the term $c^*(G)$ removes the prior preference for goals that are cheaper to achieve, and $\pi_{\max}(G)$ inserts the desired external prior over candidate goals.

Alignment with cost-based GR models. The cost-based Boltzmann formulation is derived following the same optimisation-based idea underlying Ramirez and Geffner (2010): lower-cost explanations contribute more strongly to the likelihood of an observation. Their model is motivated from a probabilistic distribution over plans together with the approximation that sums over compatible plans are dominated by their largest terms. The resulting likelihood takes the sigmoid form $P(y^* | G) = \alpha \frac{1}{1 + e^{-\beta(c^*(\neg y^*, G) - c^*(y^*, G))}}$, where $c^*(y^*, G)$ is the cost of the cheapest explanation complying with the observation y^* , and $c^*(\neg y^*, G)$ is the cost of the cheapest explanation achieving G without complying with all components of y^* .

Our derivation within a max-product inference regime instead yields a cost-gap likelihood relative to the goal-conditioned optimum $c^*(G)$, rather than a sigmoid comparison against $\neg y^*$:

$$\rho_{\max}(y^* | G) = e^{-\beta(c^*(y^*, G) - c^*(G))}.$$

This cost difference aligns more directly with the interpretation later adopted by Masters and Sardiña (2017), but in a more general GR setting. When this likelihood is combined with a uniform or externally specified prior over goals, it yields the cost-difference posterior derived above. Moreover, applying a softmax normalisation to the max-product scores gives a non-sigmoid posterior over goals of the same form advocated by Masters and Sardiña (2021). They show that the sigmoid transformation used in earlier cost-difference formulations can produce counterintuitive probability shifts, particularly under lower-rationality observations, which do not occur with the non-sigmoid formulation.

The modified joint-style score $\tilde{\rho}_{\max}(z, y^*, G)$ introduced above also makes explicit that the model contains two distinct terms: a cost-gap term over outcomes and a prior term over goals. The cost-gap term scores how far an observed outcome deviates from optimal goal-conditioned behaviour, while the prior term represents the observer’s beliefs over candidate goals. Thus, apparent divergence from an expected optimal explanation need not be attributed only to irrationality from β ; it may also arise because the observer’s prior beliefs over the acting agent’s possible goals are mistaken or incomplete.

More generally, the same derivation is not tied to plan cost specifically. Any score function over outcomes, where smaller values indicate better explanations, can be substituted for c , yielding analogous max-product likelihood and posterior scores. Plan cost is therefore one instantiation of a broader optimisation-based compatibility measure between observations and goal-conditioned outcomes.

Zero-temperature limit. As $\beta \rightarrow \infty$, the Boltzmann posterior concentrates on goals with minimal observation-constrained cost. For a set of explanations $\mathcal{Z}$ with costs $c(z)$, we use the standard asymptotic

$$\sum_{z \in \mathcal{Z}} e^{-\beta c(z)} \approx k e^{-\beta c^*},$$

where $c^* = \min_z c(z)$ and $k = |\{z \mid c(z) = c^*\}|$ is the number of optimal explanations. Applying this approximation to the observation-constrained goal scores yields

$$\rho_{\beta}(G | y^*) \approx \frac{k(y^*, G)}{k_{\min}(y^*)} e^{-\beta(c^*(y^*, G) - \min_{G'} c^*(y^*, G'))},$$

where $k(y^*, G)$ counts the optimal explanations satisfying both $z \sim y^*$ and $z \models G$, and $k_{\min}(y^*)$ counts the optimal explanations attaining the globally minimal constrained cost across goals. The exponential term dominates as $\beta \rightarrow \infty$, so posterior inference is governed primarily by minimising $c^*(y^*, G)$, with the multiplicity ratio $\frac{k_G(y^*)}{k_G}$ acting only as a secondary tie-breaking factor.

Corollary 4. As $\beta \rightarrow \infty$, Boltzmann sum-product inference converges asymptotically to a max-product inference regime in which goals are scored by optimal explanations.

In this limit, sum-product inference over explanations reduces to a max-product formulation in which each goal is scored by its best explanation, with an additional multiplicative factor tied to the number of optimal explanations. This factor is dominated by the exponential cost term and only affects the result in the case of cost ties, so that the two formulations align asymptotically as β becomes large.

The asymptotic correspondence between probabilistic and optimisation-based formulations under increasingly rational regimes is consistent with broader links between probability and optimisation explored in qualitative probabilistic reasoning (Goldschmidt and Pearl 1996) and adopted by Ramirez and Geffner (2010) which we previously discussed, while further highlighting the discrepancies between optimisation-based approximations and true outcome probabilities in lower-rationality regimes (smaller β).

Related work

Several of the presented identities may appear familiar from probabilistic models used in related inference settings, reflecting their common basis in Bayesian reasoning over latent explanations. We therefore briefly discuss the relationship to the present framework of related models.

Bayesian classification. Ordinary Bayesian classification arises as the special case in which observations fully determine the underlying outcome, i.e., $\mathcal{Z} = \mathcal{Y}$. The present GR framework generalises this setting to partially observed inference, where observations constrain the underlying outcome without uniquely determining it. Inference is therefore performed over sets of latent outcomes consistent with the observation. Observations may then correspond to incomplete feature assignments, partial trajectory information, or progressively refined information over time, with suitable projection maps ensuring that refined and coarser observations remain compatible as descriptions of the same underlying outcomes.

MAP inference. Maximum a posteriori (MAP) inference in probabilistic models selects the highest-probability latent explanation given an observation (Pearl 1988; Koller and Friedman 2009), $h^* = \arg \max_h P(h \mid x) = \arg \max_h P(x, h)$. Thus, MAP is usually posed as an optimisation query over an ordinary probabilistic model. When the underlying model admits a suitable factorisation, however, the same optimisation can be fully implemented in max-product form, propagating best-scoring partial explanations rather than summing over alternatives (Koller and Friedman 2009). The well-known Viterbi algorithm is one such instantiation for chain-structured models, computing the best hidden-state sequence by dynamic programming (Viterbi 1967).

Our max-product formulation uses a related optimisation-based structure for goal recognition: outcomes or completions play the role of latent explanations inducing observations, while candidate goals are the hypotheses being scored. Thus, the connection is structural rather than identical in inference target. MAP inference typically selects the most likely latent explanation, whereas goal recognition evaluates which higher-level goal best explains the observed behaviour through the explanations it admits.

Maximum entropy IRL. MaxEnt-IRL (Ziebart et al. 2008) provides a related probabilistic account of behaviour, but addresses a different inference problem. It models demonstrated trajectory histories with a distribution in which lower-cost trajectories are exponentially preferred, while maximum entropy avoids preferences not supported by the data. The learned object is a latent cost or reward function that makes these histories likely.

Goal recognition instead typically assumes candidate goals and uses partial observations to score which goal best explains the agent’s behaviour. In planning-based goal recognition, the domain model and goals are usually fixed, and scoring often collapses trajectory-level reasoning to dominant or optimal explanations rather than learning the objective model itself. Our identities clarify the relation-

ship between distributional reasoning over trajectories and optimisation-based goal scoring within goal recognition: on one side, they recover the cost-based interpretation of goals as supported by low-cost explanations; on the other, the outcome-distribution view allows likelihoods and posteriors to be expressed as abstract compatibility measures that support consistent Bayesian-style updates from new observations. This can facilitate both theoretical and practical connections between work pertaining to GR and MaxEnt-IRL.

Conclusion

We presented a distributional view of goal recognition through identities relating likelihoods, posteriors, and divergences between observation-induced and goal-intended distributions. The framework also yields *max-product* analogues that clarify the relation between probabilistic and optimisation-based formulations, and extend the view to score-based optimisation settings. This connects classical cost-gap interpretations with distributional compatibility measures between observations and goals. Overall, the proposed perspective provides a unified, abstract and interpretable account of goal recognition, strengthening its connections to related inference problems.

Proofs

Proof of Theorem 1. The likelihood is

$$\rho(y^* \mid G) = \sum_{z \in \mathcal{Z}} \rho(y^* \mid z, G) \rho(z \mid G).$$

By Bayes,

$$\rho(z \mid y^*, G) = \frac{\rho(y^* \mid z, G) \rho(z \mid G)}{\rho(y^* \mid G)}.$$

Hence

$$\begin{aligned} D_{\text{KL}}(\rho(z \mid y^*, G) \parallel \rho(z \mid G)) &= \sum_z \rho(z \mid y^*, G) \log \frac{\rho(z \mid y^*, G)}{\rho(z \mid G)} \\ &= \sum_z \rho(z \mid y^*, G) \log \frac{\rho(y^* \mid z, G)}{\rho(y^* \mid G)} \\ &= \sum_z \rho(z \mid y^*, G) \log \rho(y^* \mid z, G) - \log \rho(y^* \mid G). \end{aligned}$$

On the support of $\rho(z \mid y^*, G)$, $z \sim y^*$, so $\rho(y^* \mid z, G) = 1$, from the same reasoning as Equation 1. Therefore the first term is zero, giving

$$D_{\text{KL}}(\rho(z \mid y^*, G) \parallel \rho(z \mid G)) = -\log \rho(y^* \mid G).$$

□

Proof of Theorem 2. By Bayes,

$$\rho(z \mid y^*, G) = \frac{\rho(G \mid z, y^*) \rho(z \mid y^*)}{\rho(G \mid y^*)}.$$

Therefore

$$\begin{aligned}
& D_{\text{KL}}(\rho(z | y^*, G) \| \rho(z | y^*)) \\
&= \sum_z \rho(z | y^*, G) \log \frac{\rho(z | y^*, G)}{\rho(z | y^*)} \\
&= \sum_z \rho(z | y^*, G) \log \frac{\rho(G | z, y^*)}{\rho(G | y^*)} \\
&= \mathbb{E}_{\rho(z|y^*, G)} [\log \rho(G | z, y^*)] - \log \rho(G | y^*).
\end{aligned}$$

Rearranging gives the result. $\square$

Proof of Theorem 3. Since the max in the max-divergence is over non-zero outcomes of $\rho_{\max}(z | y^*, G)$, and $\rho_{\max}(y^* | z, G) = 1$ if $\rho_{\max}(z | y^*, G) > 0$:

$$\begin{aligned}
& D_{\max}(\rho_{\max}(z | y^*, G) \| \rho_{\max}(z | G)) \\
&= \max_{z: \rho_{\max}(z|y^*, G) > 0} \log \frac{\rho_{\max}(z | y^*, G)}{\rho_{\max}(z | G)} \\
&= \max_{z: \rho_{\max}(z|y^*, G) > 0} \log \frac{\rho_{\max}(y^* | z, G)}{\rho_{\max}(y^* | G)} \\
&= \max_{z: \rho_{\max}(z|y^*, G) > 0} \log \rho_{\max}(y^* | z, G) - \log \rho_{\max}(y^* | G) \\
&= -\log \rho_{\max}(y^* | G).
\end{aligned}$$

If all outcomes $z : \rho_{\max}(z | y^*, G) = 0$, then $\rho_{\max}(z | y^*, G)$ is undefined; by convention $\rho_{\max}(y^* | G) = 0$. $\square$

Proof of Theorem 4. Using max-product Bayes,

$$\rho_{\max}(z | y^*, G) \rho_{\max}(G | y^*) = \rho_{\max}(G | z, y^*) \rho_{\max}(z | y^*).$$

Thus,

$$\begin{aligned}
& D_{\max}(\rho_{\max}(z | y^*, G) \| \rho_{\max}(z | y^*)) \\
&= \max_z \log \frac{\rho_{\max}(G | z, y^*)}{\rho_{\max}(G | y^*)} \\
&= \max_z \log \rho_{\max}(G | z, y^*) - \log \rho_{\max}(G | y^*).
\end{aligned}$$

Rearranging gives

$$-\log \rho_{\max}(G | y^*) = D_{\max} - \max_z \log \rho_{\max}(G | z, y^*).$$

Since $-\max_z \log a_z = \min_z [-\log a_z]$, the result follows. $\square$

Proof of Corollary 1. By max-marginalising the goal-normalised joint score, we obtain

$$\begin{aligned}
\rho_{\max}(y^* | G) &= \max_z \mathbf{1}[z \sim y^*] \mathbf{1}[z \models G] e^{-\beta(c(z) - c^*(G))} \\
&= \max_{z: z \sim y^*, z \models G} e^{-\beta(c(z) - c^*(G))} \\
&= e^{-\beta(c^*(y^*, G) - c^*(G))}.
\end{aligned}$$

$\square$

Proof of Corollary 2. By max-product Bayes rule and Corollary 1,

$$\rho_{\max}(G | y^*) = \frac{e^{-\beta \Delta(G, y^*)} \pi_{\max}(G)}{\max_{G'} e^{-\beta \Delta(G', y^*)} \pi_{\max}(G')}.$$

$\square$

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